

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Supplementary Examination – Summer 2022

Course :B.Tech.

Branch : Civil Engg.

Semester : VI

Subject Code & Name : BTCVC602 Foundation Engineering

Max Marks : 60

Date : / /

Duration: 3.45 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the Following		
A) Explain Wash boring in detail.	CO1	06
B) Enlist different soil exploration methods & explain any one in details	CO1	06
C) Calculate of depth of soft strata underlain by hard strata use the following details : $V_1 = 600$ m/s and $V_2 = 3000$ m/s and the break in the plot was located at 60m.	CO1	06
Q.2 Solve Any Two of the Following		
A) Discuss different types of shear failure	CO3	06
B) Explain effect of position of water table on bearing capacity	CO3	06
C) A strip footing 1 m wide at its base is located at a depth of 0.8m below the ground surface. The properties of the foundation soil $\gamma = 18$ KN/m ³ $C = 30$ KN/m ³ and $\phi = 20^\circ$. Determine the safe bearing capacity using F.O.S. = 3 use. Terzaghi's analysis assume that soil fails local shear	CO3	06
Q.3 Solve Any Two of the Following		
A) What are problems associated with expansive soil	CO4	06
B) Define mat foundation & when it is preferred	CO3	06
C) Discuss design of conventional method of trapezoidal footing.	CO3	06
Q.4 Solve Any Two of the Following		
A) Discuss Negative skin friction	CO3	06
B) Explain Caisson disease in short	CO3	06
C) A wooden pile is being driven with a drop hammer weighing 20 KN and having a free fall of 1.0 m. The penetration in the last blow is 5mm. determine the load carrying capacity of the pile according to the Engineering News formula.	CO3	06
Q.5 Solve Any Two of the Following		
A) Define a) Slope failure b) Face failure c) Toe failure d) Base failure	CO2	06
B) Explain the Swedish circle method for analysis of slope	CO2	06
C) Discuss causes of slope failure	CO2	06